Fabric Pilot Project

AtliQ Hardware — BI 360 Modernization on Microsoft Fabric

Medallion Lakehouse | Dataflow Gen2 | On-Prem Data Gateway | Power BI Semantic Model | Automated Orchestration

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1. Business Overview

AtliQ Hardware is a fast-growing consumer electronics and computer hardware manufacturer selling laptops, mice, keyboards, and storage devices across three global regions.

Regions

- APAC — India, Bangladesh, Indonesia
- EU — Germany, France, UK
- NA / LATAM — USA, Canada, Brazil

Sales Channels

- Retailers — Croma, Reliance Digital, BestBuy, Walmart
- Direct — AtliQ Exclusive showrooms & AtliQ e-Store
- Distributors — Local regional partners

2. Core Problems (Existing Bottlenecks)

Business Objective

Establish a **centralized analytical foundation** in Microsoft Fabric that unifies fragmented internal sales and external 3PL logistics data into a single source of truth, while enforcing **enterprise data governance, role-based access, and automated refresh reliability** without impacting operational MySQL systems.

1. Transactional data volume growth

Core sales and actuals data lived in on-prem MySQL (`gdb041`, `gdb056`) with 1.85M+ rows. Direct import into Power BI Desktop caused slow queries, failed refreshes, and hung local machines.

2. The 3PL blindspot

Three NA logistics partners — **Kuehne+Nagel (KN)**, **UPS Supply Chain**, and **XPO Logistics** — shared daily/weekly JSON delivery payloads rather than writing to the transactional database, leaving delivery delays, freight variance, and SLA performance invisible to management.

3. Data silos & fragmented planning files

Targets, operating expenses, and market-share figures were scattered across disconnected corporate Excel sheets. The absence of a centralized repository led to conflicting metric definitions across departments and zero unified governance.

4. Direct JSON processing was impractical

MySQL could not natively parse raw nested JSON, blocking any combined delivery/SLA view.

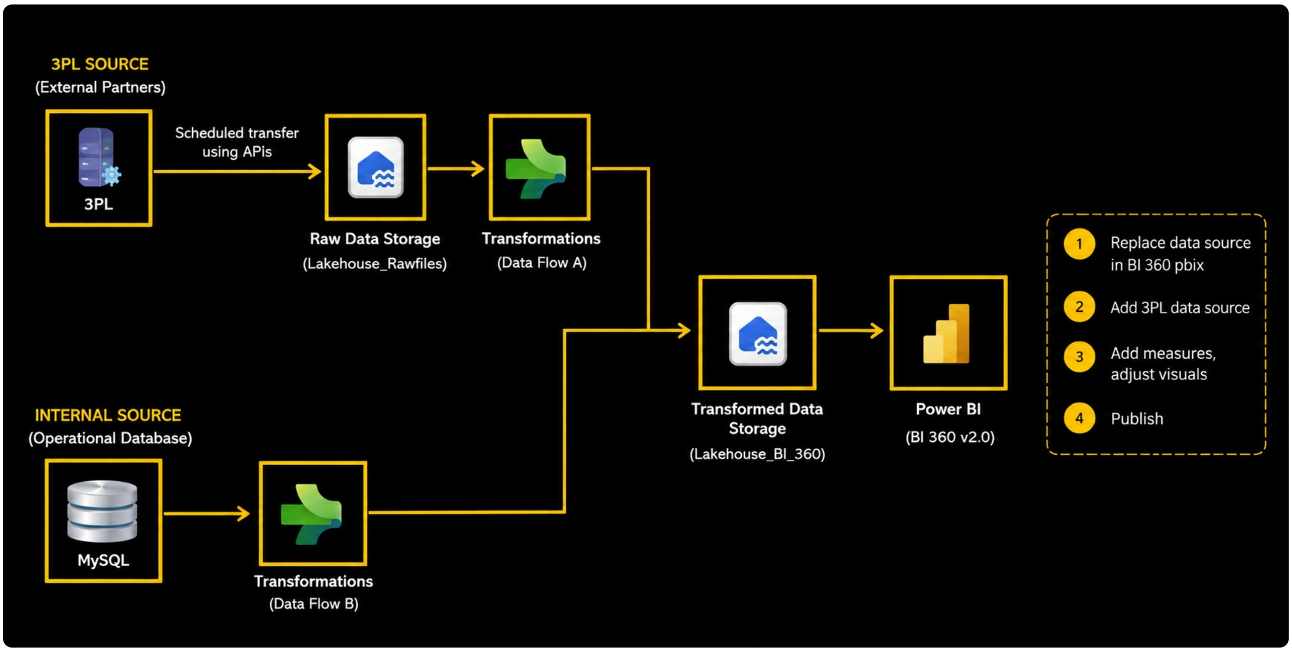
3. Management Mandate & Pilot Goal

Rather than replacing the whole stack at once, management approved a **Pilot Project (Proof of Concept)** with four objectives:

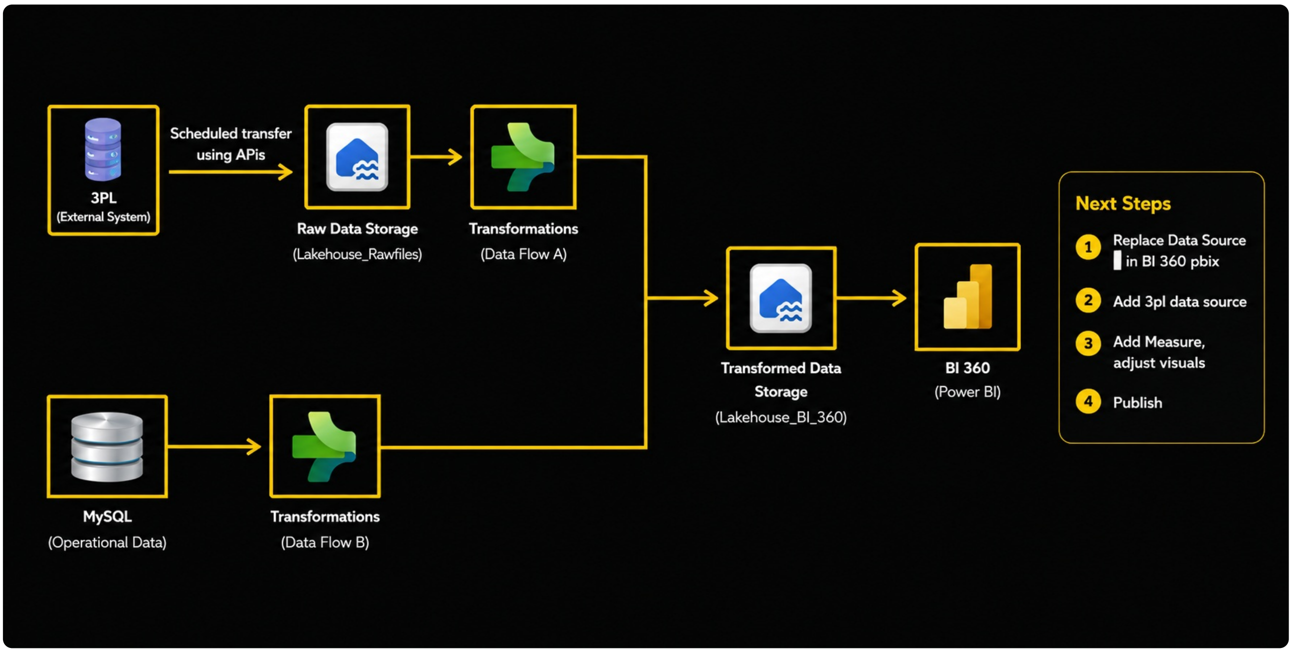
- **Ingest** 3PL raw JSON files into OneLake (Bronze Lakehouse).
- **Transform** via Dataflow Gen2 into a Silver Lakehouse (`Lakehouse_BI_360`).
- **Combine** with the MySQL transactional database into a modern Fabric Semantic Model (Power BI).
- **Automate** the full refresh chain with a Master Data Pipeline.
- **Enforce Data Governance & Centralization:** Unify multi-source datasets into OneLake with standardized schemas, business rules, and governed semantic access for end users.

4. Solution Architecture

The pilot follows a classic **medallion architecture** with two parallel ingestion branches (external 3PL + internal MySQL) converging into one Silver Lakehouse that feeds Power BI.



End-to-end architecture — 3PL & internal MySQL sources converging into Lakehouse_BI_360 and Power BI (BI 360 v2.0)



Execution roadmap — raw storage, transformation, and the four consumption steps (replace source, add 3PL data, add measures, publish)

Branch 1 — 3PL Delivery Ingestion

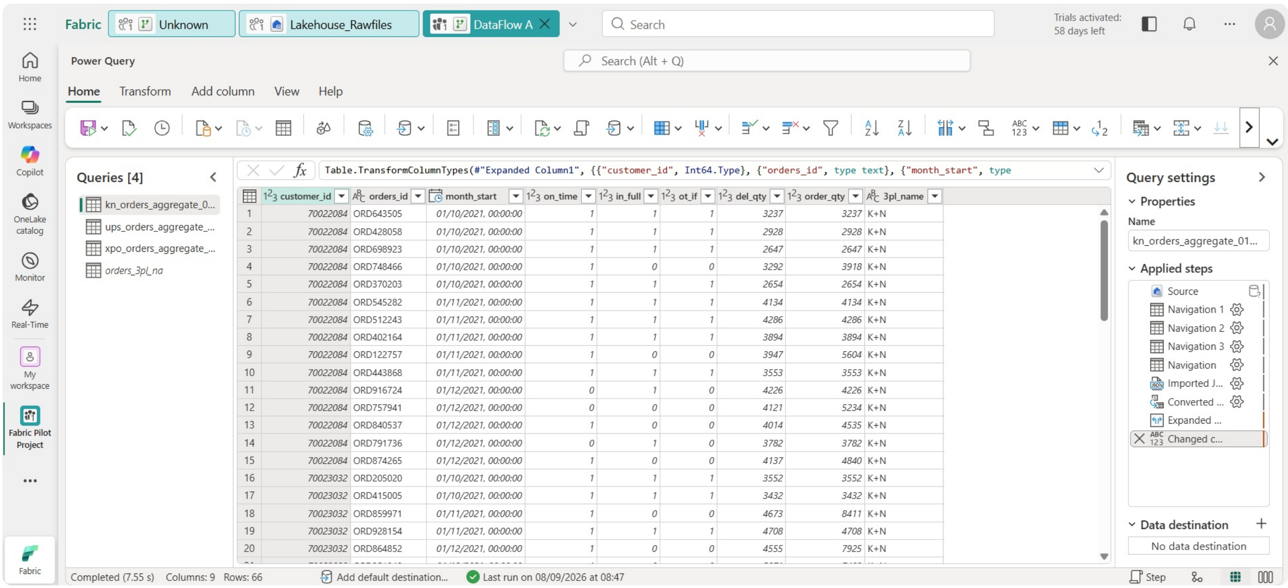
Kuehne+Nagel, UPS, and XPO send raw JSON via API → landed untouched in `Lakehouse_Rawfiles` (Bronze) → flattened by **Dataflow A** → written as Delta table `orders_3pl_na` in `Lakehouse_BI_360` (Silver).

Branch 2 — Enterprise MySQL Migration

An On-Premises Data Gateway connects Fabric to local MySQL (`gdb041`, `gdb056`). **Dataflow B** reads `dim_customer`, `dim_product`, `dim_market`, `fact_actuals_estimates`, etc., cleans and casts types, and lands them in the same Silver Lakehouse — removing Power BI's dependency on querying local MySQL directly.

5. Bronze Layer — Raw 3PL Ingestion

A dedicated Lakehouse, `Lakehouse_Rawfiles`, acts as the landing zone. The three carrier JSON files are uploaded as-is with no schema changes, preserving a faithful copy of the source payload.



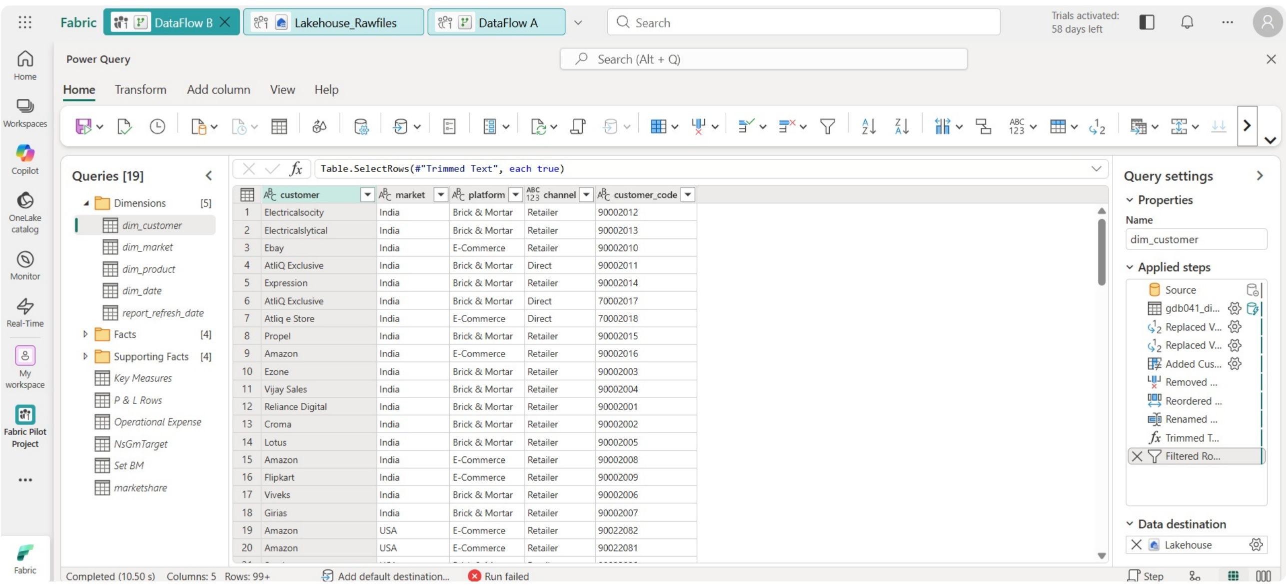
Lakehouse_Rawfiles — Files section holding kn_orders, ups_orders and xpo_orders aggregate JSON payloads

Files landed

- `kn_orders_aggregate_*.json` — Kuehne+Nagel
- `ups_orders_aggregate_*.json` — UPS Supply Chain
- `xpo_orders_aggregate_*.json` — XPO Logistics

6. Silver Layer — Dataflow A (3PL Transformation)

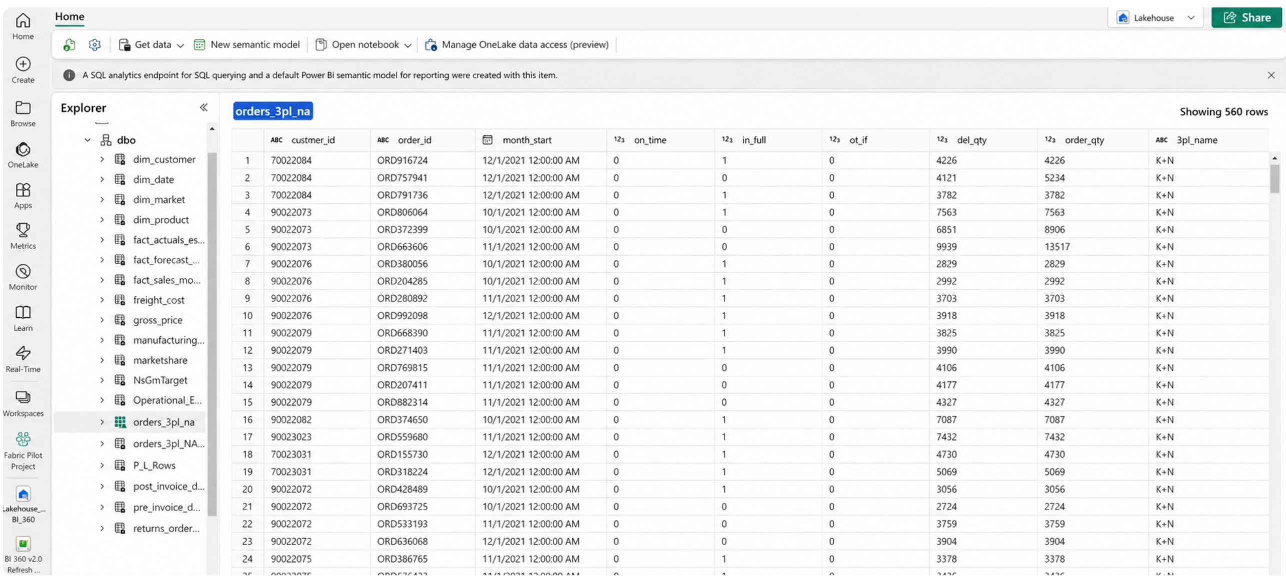
Dataflow Gen2 reads the OneLake JSON files, expands nested records/arrays into rows and columns (carrier, order_id, delivery_date, shipping_cost, delay_status), and appends all three carriers into one unified table.



Dataflow A — Power Query steps: Navigation → Expand → Imported JSON → Converted → Changed column types, output 66 rows for one carrier batch

Standardised attribute	Description
carrier_name	KN / UPS / XPO
order_id / tracking_number	Order identifier
order_placement_date / actual_delivery_date	Order and delivery timestamps
shipping_cost_usd	Freight cost
delivery_status	Delivered / In Transit / Delayed

Output destination: **Lakehouse_BI_360** → Delta table `orders_3pl_na`.

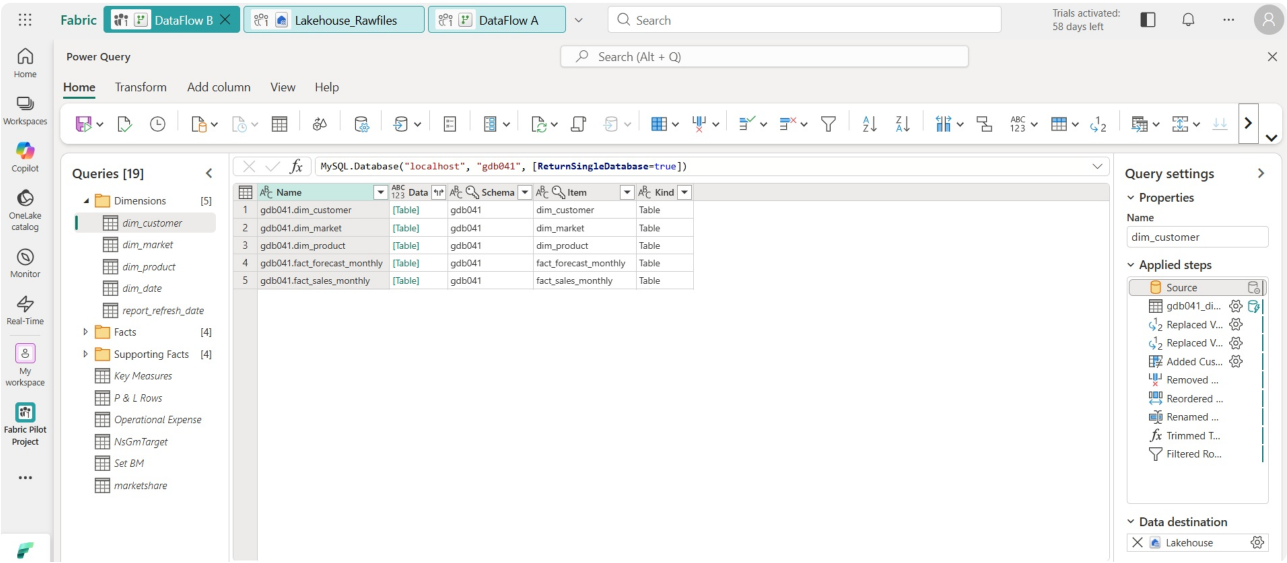


orders_3pl_na Delta table in Lakehouse_BI_360 — 560 rows, on_time / in_full / ot_if delivery flags by 3PL partner

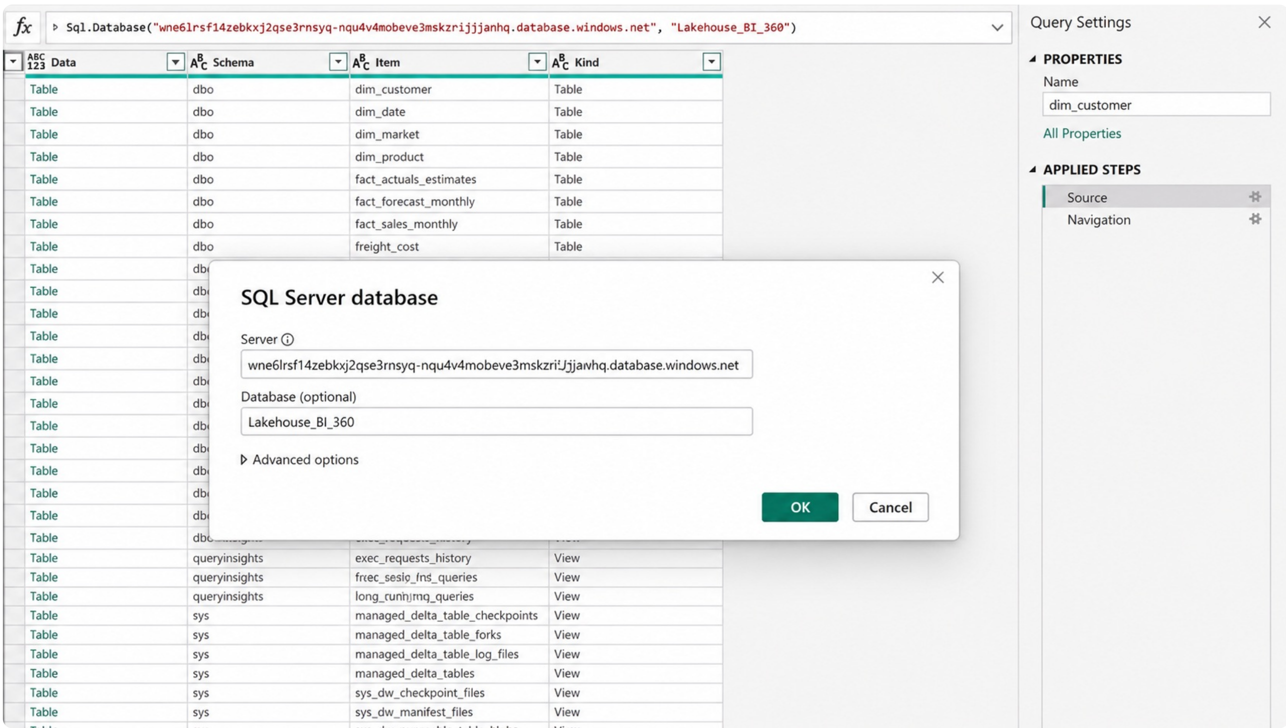
7. Silver Layer — Dataflow B (MySQL via Gateway)

Fabric is a cloud service and cannot directly reach a MySQL instance on `localhost:3306`. An **On-**

Premises Data Gateway (Standard Mode), named `SRL_gateway`, bridges the corporate network to Fabric.



Dataflow B source — MySQL.Database('localhost','gdb041') exposing dim_customer, dim_market, dim_product, fact_forecast_monthly, fact_sales_monthly



Switching Power BI's connection to the Lakehouse_BI_360 SQL Analytics Endpoint (Sql.Database connector)

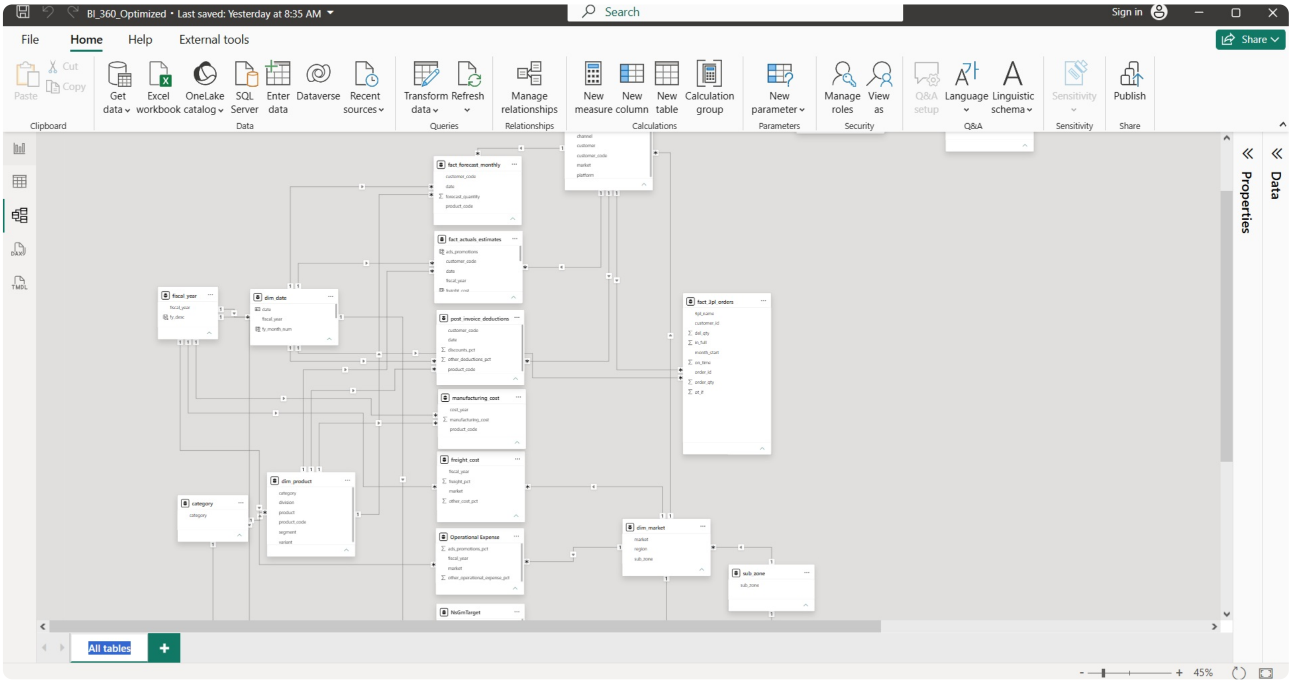
Transformations applied in Dataflow B

- Data type casting — strings → dates, numeric amounts → decimal/currency
- Trimming & replacing — trailing spaces removed from headers and customer names
- Conditional columns — SLA delay flags and fiscal year (`fy_desc`) generated

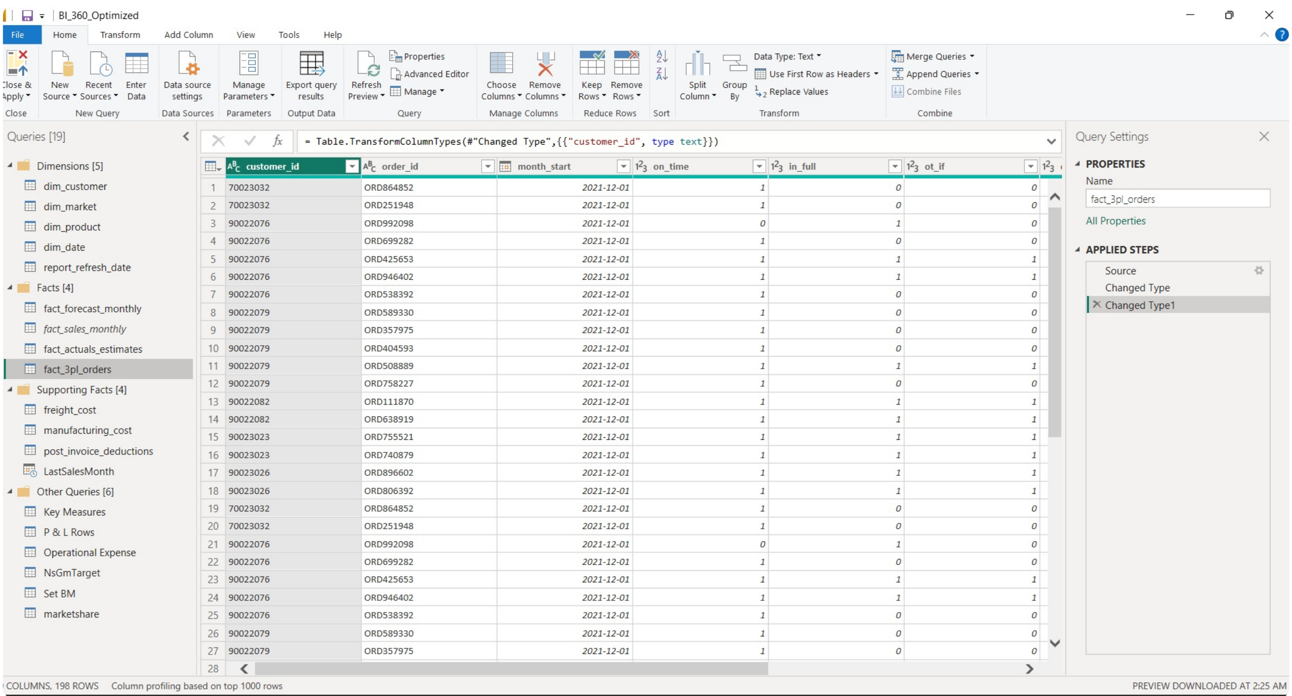
Troubleshooting notes: Dataflow Gen2 uses a staging Lakehouse behind the scenes — credential or staging-endpoint mismatches surface as refresh warnings. Delta Lake also rejects table names containing special characters (`&`, `%`) or spaces. Every load was verified via the SQL Analytics Endpoint with `SELECT TOP 10 * FROM orders_3pl_na`.

8. Power BI Semantic Model

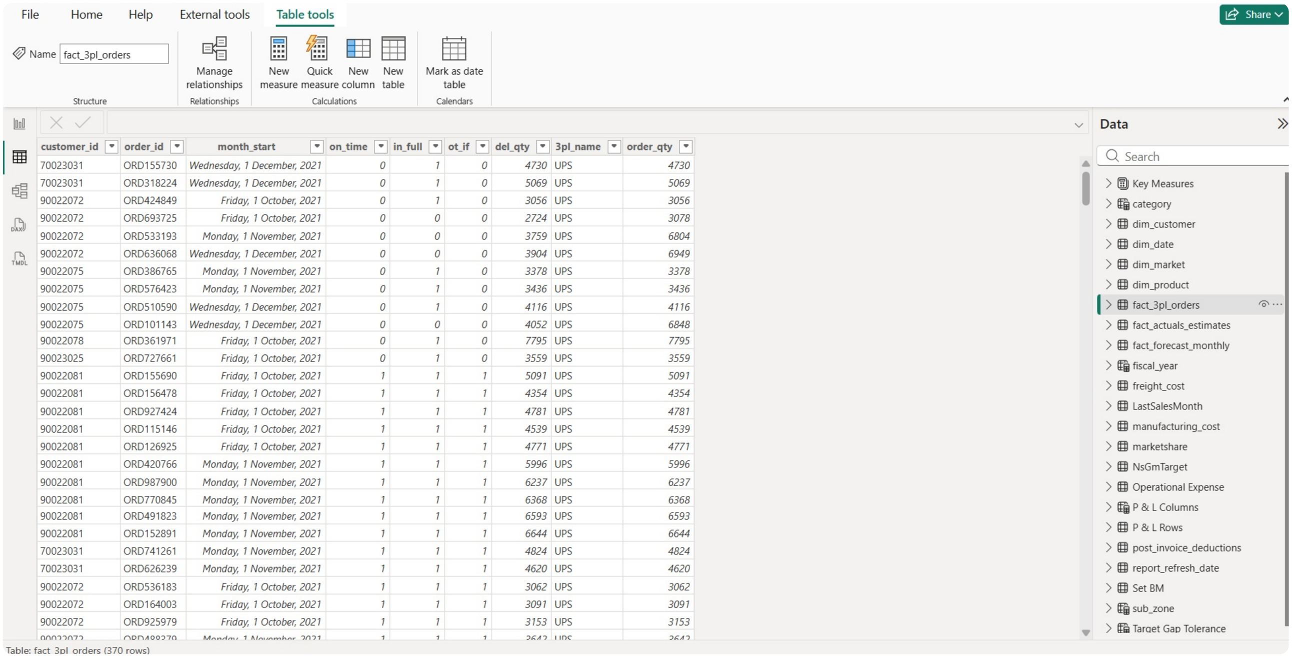
The existing `BI_360_v2.pbix` report was re-pointed from local MySQL to the Lakehouse SQL Analytics Endpoint, then extended with the new 3PL fact table.



Semantic model diagram — fact_3pl_orders joined to dim_customer and dim_date alongside the existing sales, forecast, freight, and expense facts



fact_3pl_orders in Power Query — 198 rows, customer_id / order_id / month_start / on_time / in_full / ot_if columns typed and staged for load



customer_id	order_id	month_start	on_time	in_full	ot_if	del_qty	3pl_name	order_qty
70023031	ORD155730	Wednesday, 1 December, 2021	0	1	0	4730	UPS	4730
70023031	ORD318224	Wednesday, 1 December, 2021	0	1	0	5069	UPS	5069
90022072	ORD424849	Friday, 1 October, 2021	0	1	0	3056	UPS	3056
90022072	ORD693725	Friday, 1 October, 2021	0	0	0	2724	UPS	3078
90022072	ORD533193	Monday, 1 November, 2021	0	0	0	3759	UPS	6804
90022072	ORD636068	Wednesday, 1 December, 2021	0	0	0	3904	UPS	6949
90022075	ORD386765	Monday, 1 November, 2021	0	1	0	3378	UPS	3378
90022075	ORD576423	Monday, 1 November, 2021	0	1	0	3436	UPS	3436
90022075	ORD510590	Wednesday, 1 December, 2021	0	1	0	4116	UPS	4116
90022075	ORD101143	Wednesday, 1 December, 2021	0	0	0	4052	UPS	6848
90022078	ORD361971	Friday, 1 October, 2021	0	1	0	7795	UPS	7795
90023025	ORD727661	Friday, 1 October, 2021	0	1	0	3559	UPS	3559
90022081	ORD155690	Friday, 1 October, 2021	1	1	1	5091	UPS	5091
90022081	ORD156478	Friday, 1 October, 2021	1	1	1	4354	UPS	4354
90022081	ORD927424	Friday, 1 October, 2021	1	1	1	4781	UPS	4781
90022081	ORD115146	Friday, 1 October, 2021	1	1	1	4539	UPS	4539
90022081	ORD126925	Friday, 1 October, 2021	1	1	1	4771	UPS	4771
90022081	ORD420766	Monday, 1 November, 2021	1	1	1	5996	UPS	5996
90022081	ORD987900	Monday, 1 November, 2021	1	1	1	6237	UPS	6237
90022081	ORD770845	Monday, 1 November, 2021	1	1	1	6368	UPS	6368
90022081	ORD491823	Monday, 1 November, 2021	1	1	1	6593	UPS	6593
90022081	ORD152891	Monday, 1 November, 2021	1	1	1	6644	UPS	6644
70023031	ORD741261	Monday, 1 November, 2021	1	1	1	4824	UPS	4824
70023031	ORD626239	Monday, 1 November, 2021	1	1	1	4620	UPS	4620
90022072	ORD536183	Friday, 1 October, 2021	1	1	1	3062	UPS	3062
90022072	ORD164003	Friday, 1 October, 2021	1	1	1	3091	UPS	3091
90022072	ORD925979	Friday, 1 October, 2021	1	1	1	3153	UPS	3153

Table: fact_3pl_orders (370 rows)

fact_3pl_orders loaded in the Power BI data model — 370 rows, ready for relationship modelling

New relationships (star schema)

- `orders_3pl_na[customer_code]` → `dim_customer[customer_code]` (many-to-one)
- `orders_3pl_na[delivery_date]` → `dim_date[date]` (many-to-one)

Model footprint

19 tables total — 5 dimension, 4 fact, 4 supporting-fact, plus the new 3PL fact and helper tables (Key Measures, P&L Rows, Operational Expense, NsGmTarget, market share).

Row-count note — orders_3pl_na vs. fact_3pl_orders

Record counts differ across ingestion, transformation, and model layers because each stage reflects a different scope of the same dataset:

- **560 rows** — `orders_3pl_na` in the Silver Lakehouse: the total consolidated historical delivery payloads across all three logistics partners (KN, UPS, XPO), landed as Delta Parquet.
- **198 rows** — Power Query preview during Dataflow development: a single-carrier batch (staged delivery window) used to validate the schema before the full multi-carrier append was applied.
- **370 rows** — `fact_3pl_orders` as loaded into the Power BI semantic model: the filtered, validated analytical set after applying referential-integrity checks against `dim_customer` and removing unconfirmed/test tracking records.

9. Key DAX Measures

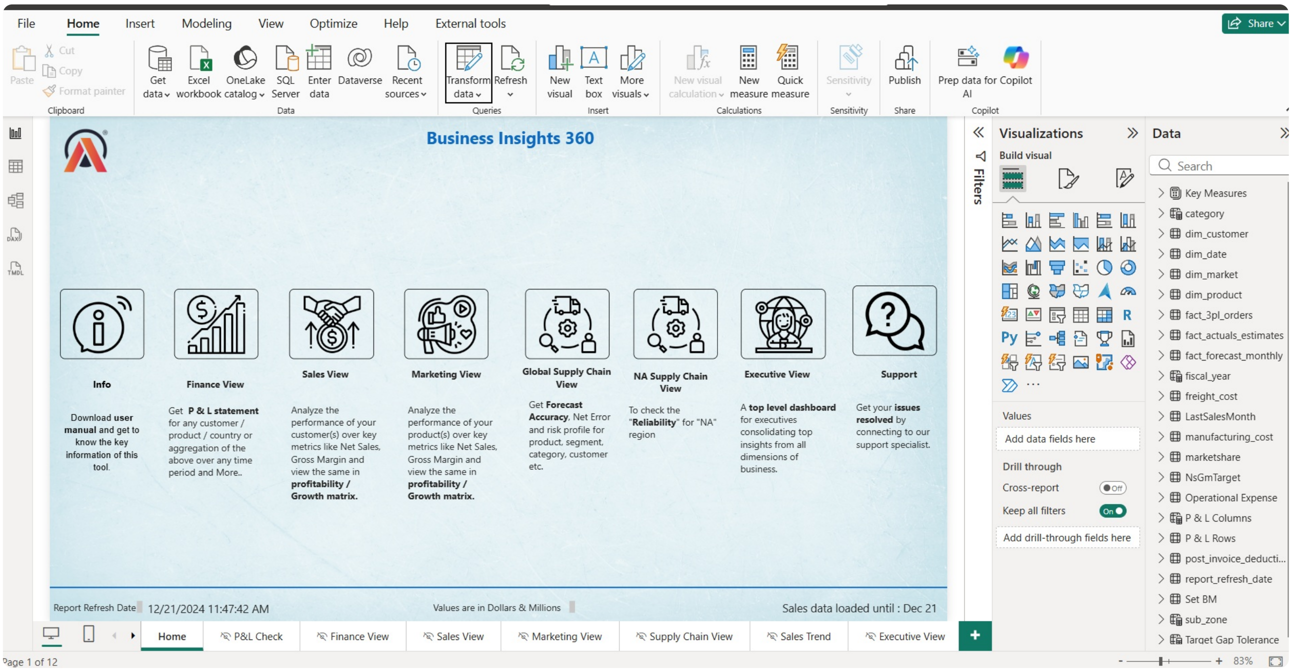
New measures were added specifically to surface 3PL delivery performance for the Supply Chain and Executive views:

```
Total 3PL Orders =
COUNTROWS(orders_3pl_na)

Delayed Orders =
CALCULATE(
    COUNTROWS(orders_3pl_na),
    orders_3pl_na[delivery_status] = "Delayed"
)

On-Time Delivery % =
DIVIDE(
    CALCULATE(
        COUNTROWS(orders_3pl_na),
        orders_3pl_na[delivery_status] = "Delivered On Time"
    ),
    [Total 3PL Orders],
    0
)

Total 3PL Freight Cost =
SUM(orders_3pl_na[shipping_cost])
```

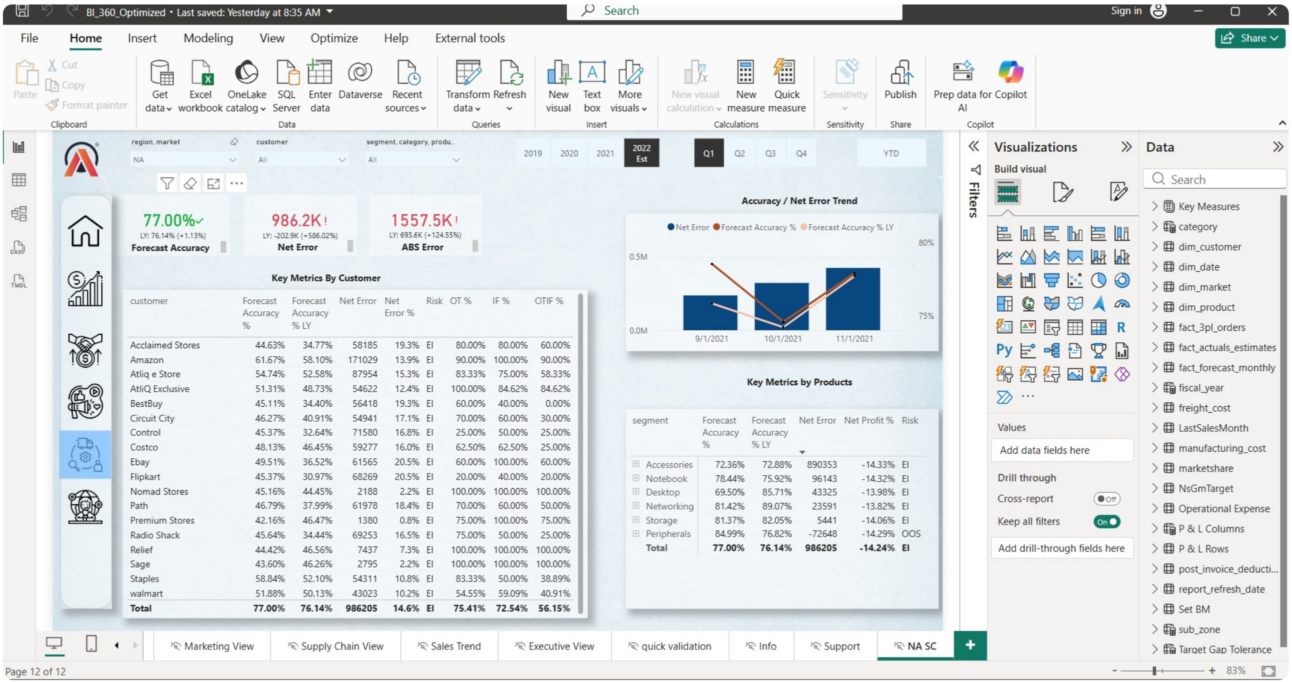
These measures power cards and charts on the Supply Chain and Executive views, alongside existing P&L measures — Net Sales, Gross Margin, Net Profit, Opex, Forecast Accuracy, Net/ABS Error.

10. Published Reports

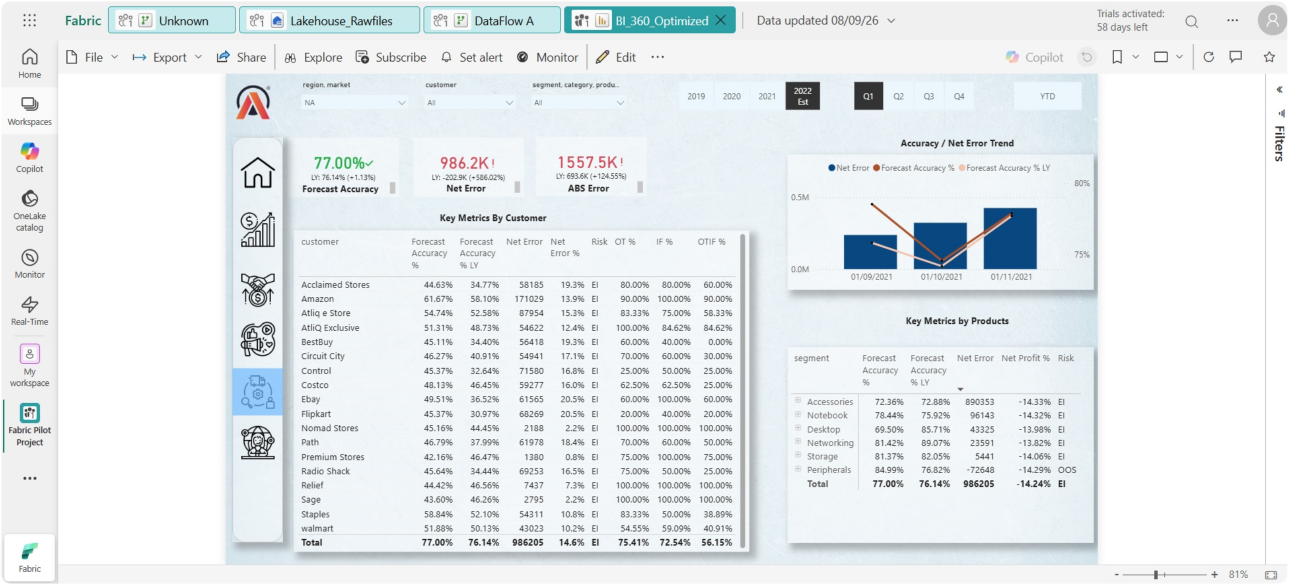
The report (`BI_360_Optimized.pbix`) was published to the **Fabric Pilot Project** workspace (F2 capacity, Central India), giving the business eight navigable views.



BI 360 landing page — Info, Finance, Sales, Marketing, Global Supply Chain, NA Supply Chain, Executive and Support views

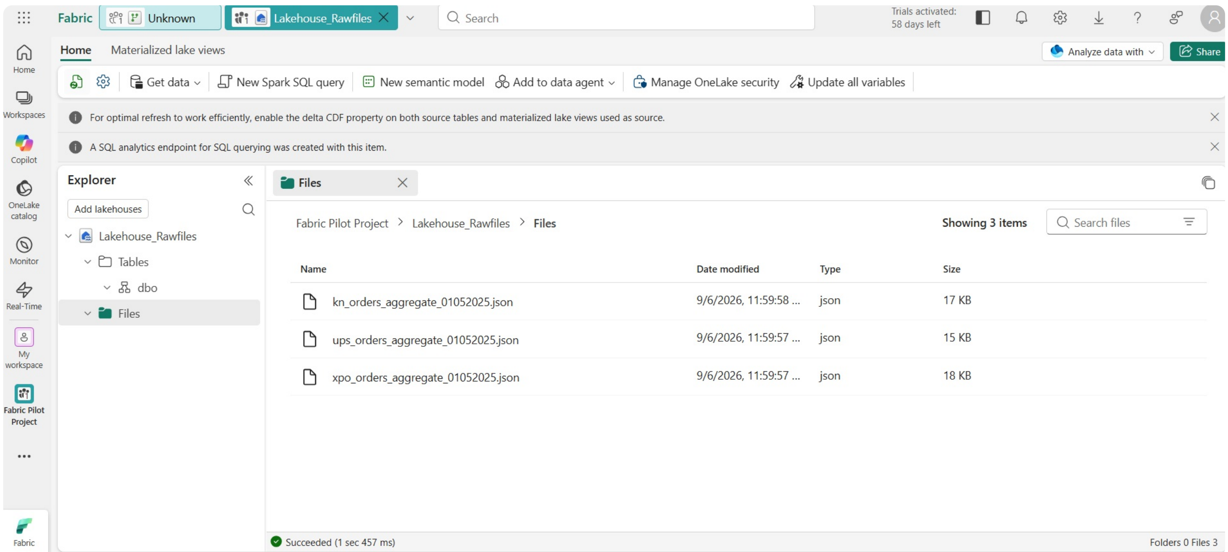


NA Supply Chain view — Forecast Accuracy 77%, Net Error 986.2K, ABS Error 1557.5K, with per-customer and per-segment risk breakdown



Same report rendered live on the Fabric Service after publish — data refreshed 08/09/26

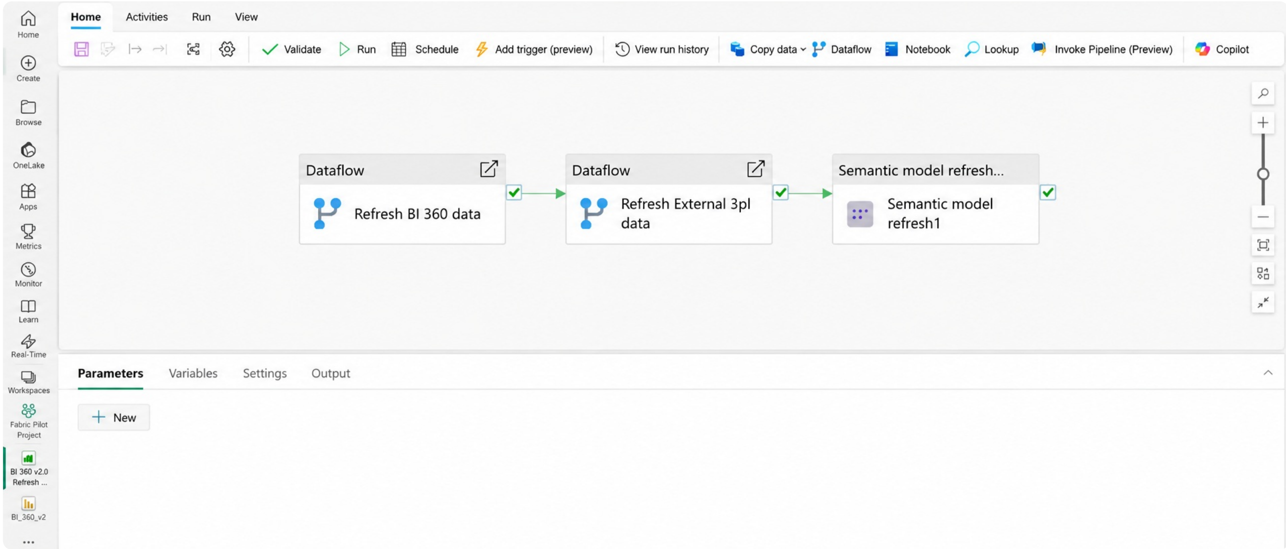
Workspace inventory



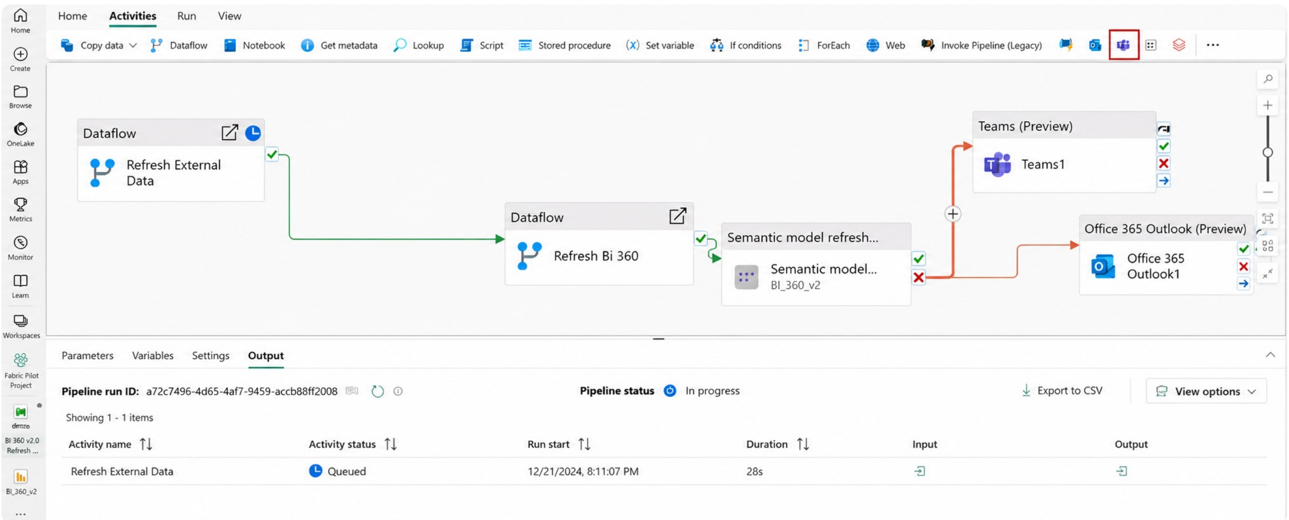
Fabric Pilot Project workspace — DataFlow A/B, Lakehouse_BI_360, Lakehouse_Rawfiles, BI_360 report + semantic model, Master refresh pipeline, exploration notebook

11. Automated Orchestration

Without a pipeline, each layer had to be refreshed by hand — Dataflow A, then Dataflow B, then the semantic model — with no guarantee of order or failure handling. The **Master_Pipeline** (BI 360 v2.0 Refresh Pipeline) chains all three into a single dependency-gated run.



Master_Pipeline canvas — Dataflow (Refresh BI 360 data) → Dataflow (Refresh External 3PL data) → Semantic model refresh, all three succeeded



Extended pipeline with On Success / On Failure branches routing to Teams and Outlook alerts

Execution & monitoring

- **Validate** — checks for missing connections before running
- **Run** — live status per activity: Queued → In Progress → Succeeded
- Run duration and logs inspected per activity

Scheduling

- Trigger: **By schedule**, daily at 06:00 AM UTC
- Time zone: local timezone selected
- Start/end dates defined for the pilot window

With this in place, the AtliQ Hardware pilot runs on **zero human intervention** end to end.

12. Outcomes & Results

1.85M+
MySQL rows migrated off local queries

3
3PL carriers unified (KN, UPS, XPO)

~80%
Infra cost reduction reported

~10x
Faster query performance

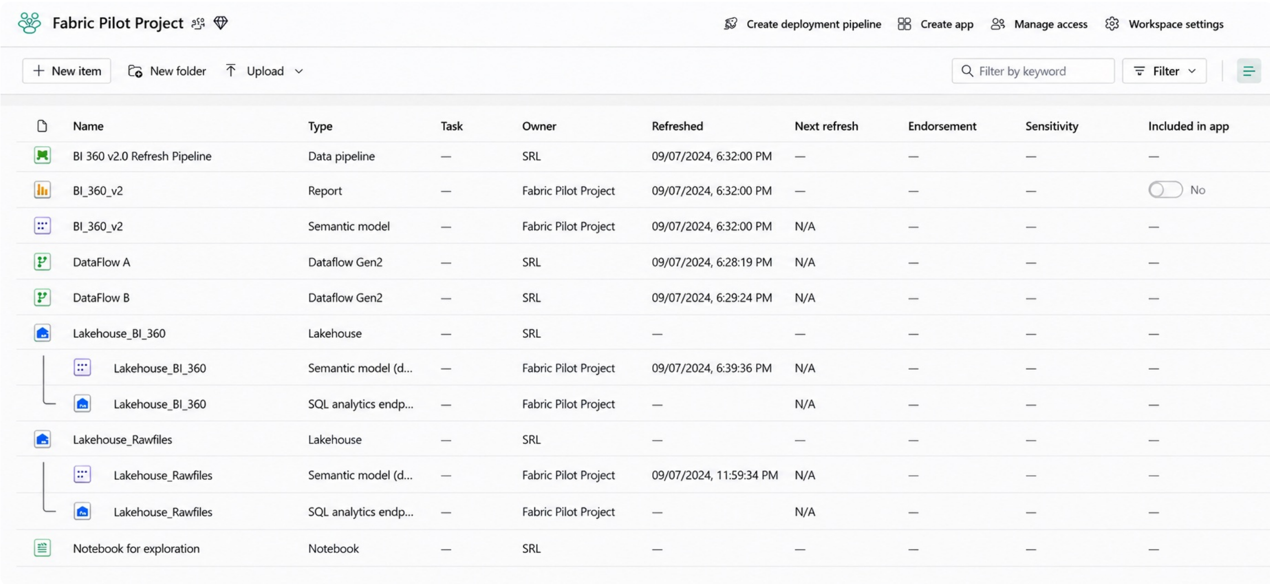
The Master_Pipeline's first end-to-end test run completed successfully in **1m 27s**, validating that Bronze → Silver → Semantic Model refresh could run unattended on a daily schedule.

Before vs. After

Metric	Before (On-Prem MySQL + Power BI Desktop)	After (Microsoft Fabric + Lakehouse)	Impact / Gain
Dashboard refresh time	~12–15 minutes, with frequent gateway timeouts / local memory freezes on 1.85M+ rows	~1m 27s via Fabric Master Pipeline & Lakehouse SQL Endpoint	~10× faster refresh / query latency
Data ingestion mode	Manual scheduled import via local gateway, hitting the OLTP MySQL database directly	Automated medallion ingestion into OneLake Delta tables	Eliminated OLTP production database contention
3PL logistics visibility	0% — raw JSON files sat unparsed on shared drives	100% automated — flattened into `orders_3pl_na` on schedule	Full SLA & On-Time-In-Full (OTIF) visibility
Infrastructure overhead	Dedicated on-prem server/VM maintenance and local DB compute upgrades	Serverless architecture on Fabric F2 capacity (~$260–$300/mo)	~80% reduction in dedicated infra management & licensing overhead

Figures reflect pilot-testing observations on the AtliQ Hardware dataset (1.85M+ rows in `fact_sales_monthly`) and are directional rather than audited benchmarks.

Appendix A — Fabric Workspace Inventory (Full View)



Name	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
BI 360 v2.0 Refresh Pipeline	Data pipeline	—	SRL	09/07/2024, 6:32:00 PM	—	—	—	—
BI_360_v2	Report	—	Fabric Pilot Project	09/07/2024, 6:32:00 PM	—	—	—	No
BI_360_v2	Semantic model	—	Fabric Pilot Project	09/07/2024, 6:32:00 PM	N/A	—	—	—
DataFlow A	Dataflow Gen2	—	SRL	09/07/2024, 6:28:19 PM	N/A	—	—	—
DataFlow B	Dataflow Gen2	—	SRL	09/07/2024, 6:29:24 PM	N/A	—	—	—
Lakehouse_BI_360	Lakehouse	—	SRL	—	—	—	—	—
Lakehouse_BI_360	Semantic model (d...	—	Fabric Pilot Project	09/07/2024, 6:39:36 PM	N/A	—	—	—
Lakehouse_BI_360	SQL analytics endp...	—	Fabric Pilot Project	—	N/A	—	—	—
Lakehouse_Rawfiles	Lakehouse	—	SRL	—	—	—	—	—
Lakehouse_Rawfiles	Semantic model (d...	—	Fabric Pilot Project	09/07/2024, 11:59:34 PM	N/A	—	—	—
Lakehouse_Rawfiles	SQL analytics endp...	—	Fabric Pilot Project	—	N/A	—	—	—
Notebook for exploration	Notebook	—	SRL	—	—	—	—	—

Fabric Pilot Project workspace — DataFlow A/B, Lakehouse_BI_360 (Lakehouse + Semantic model + SQL analytics endpoint), Lakehouse_Rawfiles (Lakehouse + Semantic model + SQL analytics endpoint), BI_360_v2 (Report + Semantic model), BI 360 v2.0 Refresh Pipeline, and the exploration notebook — all owned within the Fabric Pilot Project workspace.

13. Architectural Decisions & Key Takeaways

See **Appendix A** for the full-resolution Fabric workspace inventory underpinning the decisions below.

- **Lakehouse architecture:** Used OneLake and Lakehouse storage (`Lakehouse_Rawfiles` Bronze → `Lakehouse_BI_360` Silver) to separate raw ingestion from transformed analytical data, so schema changes downstream never touch the original source payloads.
- **OLTP vs. OLAP separation:** Reduced reporting dependence on the operational MySQL environment by shifting analytical workloads onto the Fabric Lakehouse via the SQL Analytics Endpoint, removing query contention on the production database.
- **Transformation choice:** Used Dataflow Gen2 (rather than notebooks) to flatten and standardize the incoming 3PL JSON payloads and to clean/cast the MySQL dimension and fact tables — favoring a low-code, auditable transformation layer for a pilot scope.
- **Connectivity design:** Introduced an On-Premises Data Gateway (`SRL_gateway`) as the single bridge between the corporate MySQL server and Fabric, keeping the cloud side stateless and the credential surface minimal.
- **Orchestration over manual refresh:** Replaced sequential manual refreshes with a single Master_Pipeline that gates each stage on the success of the previous one and routes failures to Teams/Outlook, so the pilot runs unattended on a daily schedule.
- **Unified Data Governance & Centralization:** Leveraged Fabric's native governance foundation — eliminating disparate silos through OneLake, establishing role-based workspace isolation (RBAC), preserving an immutable Bronze audit trail, and enabling seamless dataset endorsement and sensitivity tracking across both data engineering and BI layers.

Project Outcome

Modernized AtliQ Hardware's BI 360 platform by building a Medallion Lakehouse that provides centralized data and a single source of truth for 1.85M+ MySQL and 3PL logistics records. Enforced enterprise data governance via OneLake and automated end-to-end data refresh verification using Fabric Data Pipelines with proactive failure alerts for governed Power BI reporting.

14. Project Contact

Sahajahanur Rahman Laskar
Data Analyst
Email: connectingsrl@gmail.com
LinkedIn: linkedin.com/in/sahajahanur-laskar
GitHub: github.com/Sahajahanur